USHA MITTAL INSTITUTE OF TECHNOLOGY

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NEWS PORTAL

(NOVA PRESS)

A Synopsis Submitted in Partial Fulfillment of the Requirements for the Degree of Bachelor in Technology in Computer Engineering

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1. Abstract

The proposed News Portal (Nova Press) is a full-stack web-based application intended to provide users with a centralized, organized, responsive, and interactive platform for accessing news articles. The system will present news through a clear interface and organize articles into different categories. Users will be able to browse news, open complete articles using the Read More facility, and interact through Like, Dislike, and Comments.

The application will be developed using modern web technologies. React.js will be used for the frontend and reusable user-interface components. Node.js and Express.js will provide backend services and REST APIs, while MongoDB will store and retrieve news and interaction data. JWT will be used for secure user authentication and authorization. Cloudinary will be used for managing uploaded news images and media. Socket.IO will support real-time notifications and updates.

An AI-based feature using the Google Gemini API is proposed to enhance the portal by assisting with news summarization and other content-support functions. Tailwind CSS will be used to create a responsive and modern user interface, and Nodemailer can be used for email notifications. The system will follow a client-server architecture in which the React frontend communicates with backend APIs for data processing and database operations.

The project aims to demonstrate practical implementation of frontend development, REST API communication, authentication, database management, media handling, real-time communication, and AI-assisted functionality in a single full-stack application.

2. Introduction

News and information are an important part of everyday life, and the Internet has become one of the primary ways through which people access current information. Users may need to move between different sources to find news related to different subjects. A centralized News Portal can provide a structured environment where articles are organized and presented systematically.

Nova Press will provide an accessible interface for browsing news through different categories. Users will be able to view article information, open complete content through Read More, and interact with articles using Like, Dislike, and Comments. Authentication will allow the system to identify users and control access to relevant functions.

From a technical perspective, the project demonstrates the development of a modern full-stack web application. React.js and Tailwind CSS will be used for the frontend, while Node.js and Express.js will provide backend services. MongoDB will act as the database. JWT will support authentication, Cloudinary will manage media, Socket.IO will enable real-time communication, and an AI service will assist with selected content-related functions.

3. Problem Statement

With the increasing availability of information on the Internet, users may have to access multiple sources to find news related to different subjects. Unorganized presentation of articles can make it difficult to browse information efficiently. A news platform therefore needs a structured interface through which articles can be categorized, displayed, searched, and accessed easily.

A modern news platform also requires user interaction and secure management of user data. A basic static website may display information but may not provide mechanisms for authentication, article reactions, discussions, notifications, or efficient media management. These requirements require a backend and database capable of securely storing and retrieving article, user, and interaction data.

Therefore, the proposed project will develop a web-based News Portal that provides organized category-based browsing, article viewing, Like, Dislike, Comments, secure authentication, media management, real-time notifications, and selected AI-assisted content functionality.

4. Objectives

- To develop a web-based platform for displaying and managing news articles.
- To organize news articles into suitable categories for easier browsing.
- To provide a Read More facility for viewing complete article content.
- To implement Like and Dislike functionality for news articles.
- To provide Comments for user interaction with articles.
- To implement secure user authentication and authorization using JWT.
- To develop the frontend using React.js and Tailwind CSS.
- To develop backend services and REST APIs using Node.js and Express.js.
- To store and retrieve news, user, and interaction data using MongoDB.
- To manage article images and media using Cloudinary.
- To provide real-time notifications or updates using Socket.IO.
- To integrate an AI service for features such as news summarization and content assistance.
- To provide email notifications where required using Nodemailer.
- To integrate all components into a functional, responsive full-stack application.

5. Related Work

Online news platforms commonly provide users with digital access to articles through web-based interfaces. Such systems generally organize content using categories and provide article pages through which users can read detailed information. Modern platforms also use database-driven architectures so that content can be stored, retrieved, and updated dynamically.

Web application development has evolved from static HTML-based pages to full-stack applications using component-based frontend frameworks, server-side APIs, database systems, authentication mechanisms, cloud media services, and real-time communication. React.js can be used for reusable interactive interfaces, while Node.js and Express.js can provide backend services. MongoDB provides document-oriented storage.

The proposed Nova Press project extends this approach by combining secure JWT authentication, cloud-based media management, real-time communication through Socket.IO, and AI-assisted content functionality. These features are proposed as part of the application rather than limiting the portal to static news display.

6. Proposed Method

The proposed system will be a full-stack News Portal consisting of a React.js and Tailwind CSS frontend, a Node.js and Express.js backend, and a MongoDB database. The frontend will provide the interface through which users browse and interact with news articles. Requests from the frontend will be sent to backend APIs, which will process operations and communicate with MongoDB.

JWT will be used to authenticate users and protect authorized operations. Cloudinary will be integrated for storing and serving article images and media. Socket.IO will provide real-time communication for selected notifications or updates. An AI service using the Google Gemini API is

proposed for features such as generating concise article summaries or assisting content preparation.

The retrieved information will be sent to the React application and rendered through appropriate interface components. Article interaction features will also be handled through communication between the frontend and backend.

Technology	Role
React.js	Frontend development and reusable UI components
Tailwind CSS	Responsive and modern interface styling
Node.js + Express.js	Backend services and REST API communication
MongoDB	Storage and retrieval of application data
JWT	Authentication and authorization
Cloudinary	News image and media management
Socket.IO	Real-time notifications and updates
Google Gemini API	AI-assisted news summarization/content support
Nodemailer	Email notification support

7. Methodology

7.1 Requirement Analysis

The requirements of Nova Press will be identified, including article display, category-based browsing, Read More functionality, Like and Dislike interaction, Comments, authentication, media management, notifications, AI-assisted features, backend operations, and database requirements.

7.2 Frontend Development

The frontend will be developed using React.js. Components will be organized so that news information and user interactions can be displayed and handled systematically. Tailwind CSS will be used for responsive styling.

7.3 Backend Development

Node.js and Express.js will be used to develop backend services and REST API endpoints through which the frontend can request and manage news-related information.

7.4 Authentication and Authorization

JWT-based authentication will be implemented for secure login and protected operations. The backend will verify authentication information before allowing authorized actions.

7.5 Database Integration

MongoDB will be integrated with the backend to store and retrieve news articles, user information, comments, reactions, and other application data.

7.6 Media Management

Cloudinary will be used to manage uploaded article images and other supported media. The application will store the required media references and display them through the frontend.

7.7 AI-Assisted Functionality

The proposed AI integration will assist with selected content-related functions, such as producing concise summaries of articles. The generated content will be handled through the backend before being presented to users.

7.8 Real-Time Communication

Socket.IO will be used where real-time communication is required, such as notifications or selected updates without requiring a full page refresh.

7.9 News Display and Categorization

News articles will be displayed through the frontend and organized according to available categories. Users will be able to browse articles and select an article to access complete content through Read More.

7.10 User Interaction

The system will provide Like, Dislike, and Comments features. These operations will allow users to interact with available news content.

7.11 API and Database Communication

The React frontend will communicate with the Express backend through API requests. The backend will process requests, communicate with MongoDB, and return the required information.

7.12 Testing and Integration

Frontend components, backend APIs, authentication, database operations, media handling, article display, AI functionality, and user interaction features will be tested and integrated as development progresses.

8. Proposed System Modules

- User Module: Registration, login, profile and authenticated interactions.
- News Module: Create, retrieve, organize, display, and view complete news articles.
- Category Module: Browse articles according to news categories.
- Interaction Module: Like, Dislike and Comments.
- Authentication Module: JWT-based login and authorization.
- Media Module: Cloudinary-based image/media management.
- AI Module: AI-assisted article summarization/content support.
- Notification Module: Real-time notifications using Socket.IO and optional email notifications using Nodemailer.
- Admin/Management Module: Manage news content, categories and application data.

9. Expected Outcome

The expected outcome is a functional, responsive, and interactive News Portal named Nova Press. The completed system will allow users to browse categorized news, read complete articles, interact through Like, Dislike and Comments, and use secure authenticated features. The application will demonstrate integration of frontend, backend, database, authentication, cloud media, real-time communication, and AI-assisted functionality.

The project is expected to provide a practical demonstration of full-stack web development and the integration of modern technologies into a single web application.

10. References

- React Documentation – React library for building user interfaces.
- Node.js Documentation – JavaScript runtime environment.
- Express.js Documentation – Node.js web application framework.
- MongoDB Documentation – Document database platform.
- MDN Web Docs – Web development technologies and JavaScript reference.
- Tailwind CSS Documentation – Utility-first CSS framework.
- JSON Web Token (JWT) documentation – Token-based authentication concept.
- Cloudinary Documentation – Cloud-based image and media management.
- Socket.IO Documentation – Real-time bidirectional communication.
- Google Gemini API documentation – Generative AI capabilities.
- Nodemailer Documentation – Email sending for Node.js applications.